AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 26 · STUDENT LESSON

Image and Voice Pipelines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 2: inspect information passed between stages.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of speech pipelines and multimodal evidence. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of speech **pipeline**s and multimodal **evidence**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 26: Image and Voice **Pipeline**s

Perception systems transform **signal**s through multiple fallible stages.



THINKER MISSION

Trace speech-to-command and image-to-label **pipeline**s.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Image generation and provenance. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can trace where speech information changes and resolve a declared-authority conflict.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Image and Voice Pipelines case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use image and voice **pipeline**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A multimodal **pipeline** may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

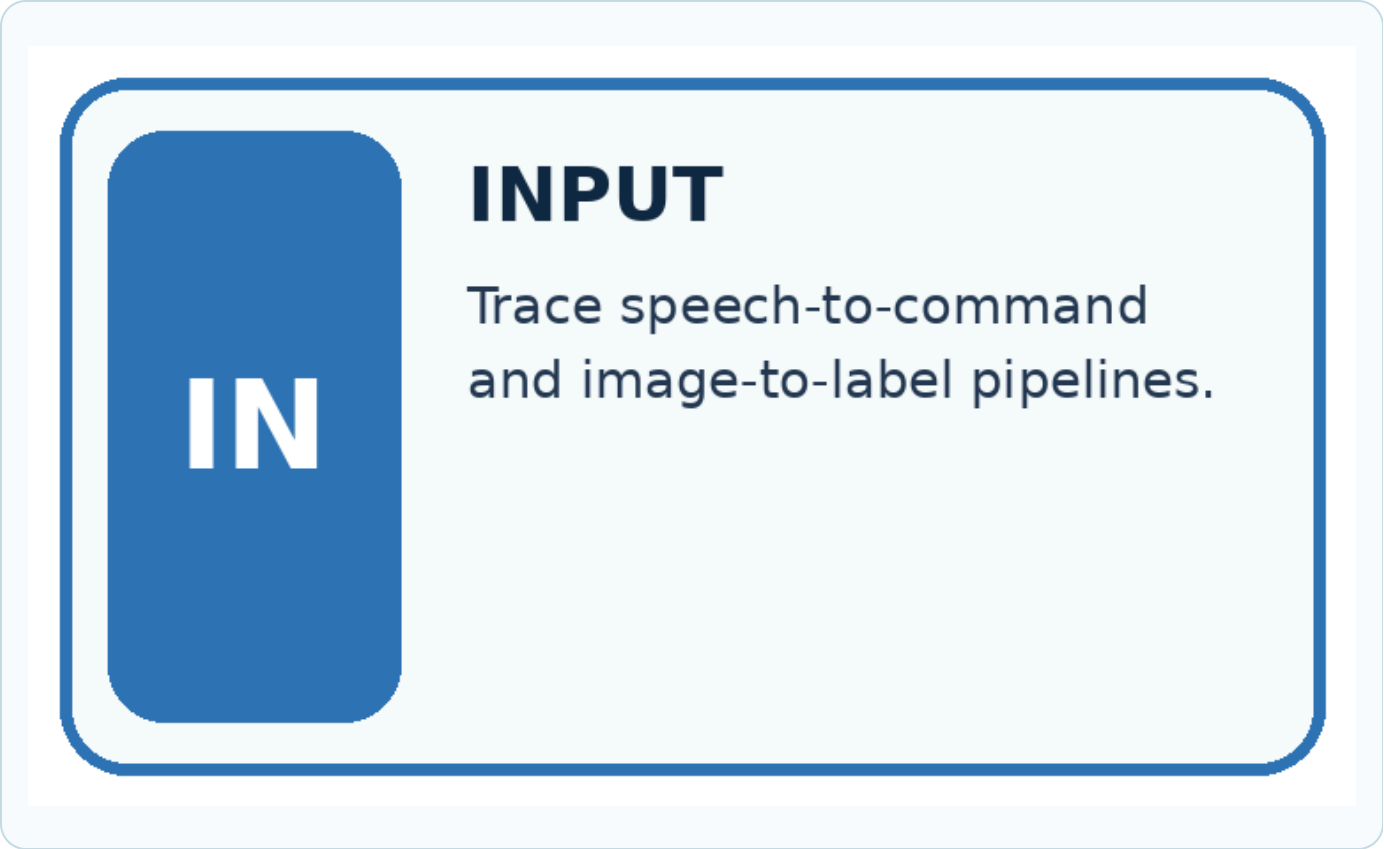
See how it works

> Follow the mechanism for Image and Voice Pipelines. Connect each labelled input, deciding step and result.

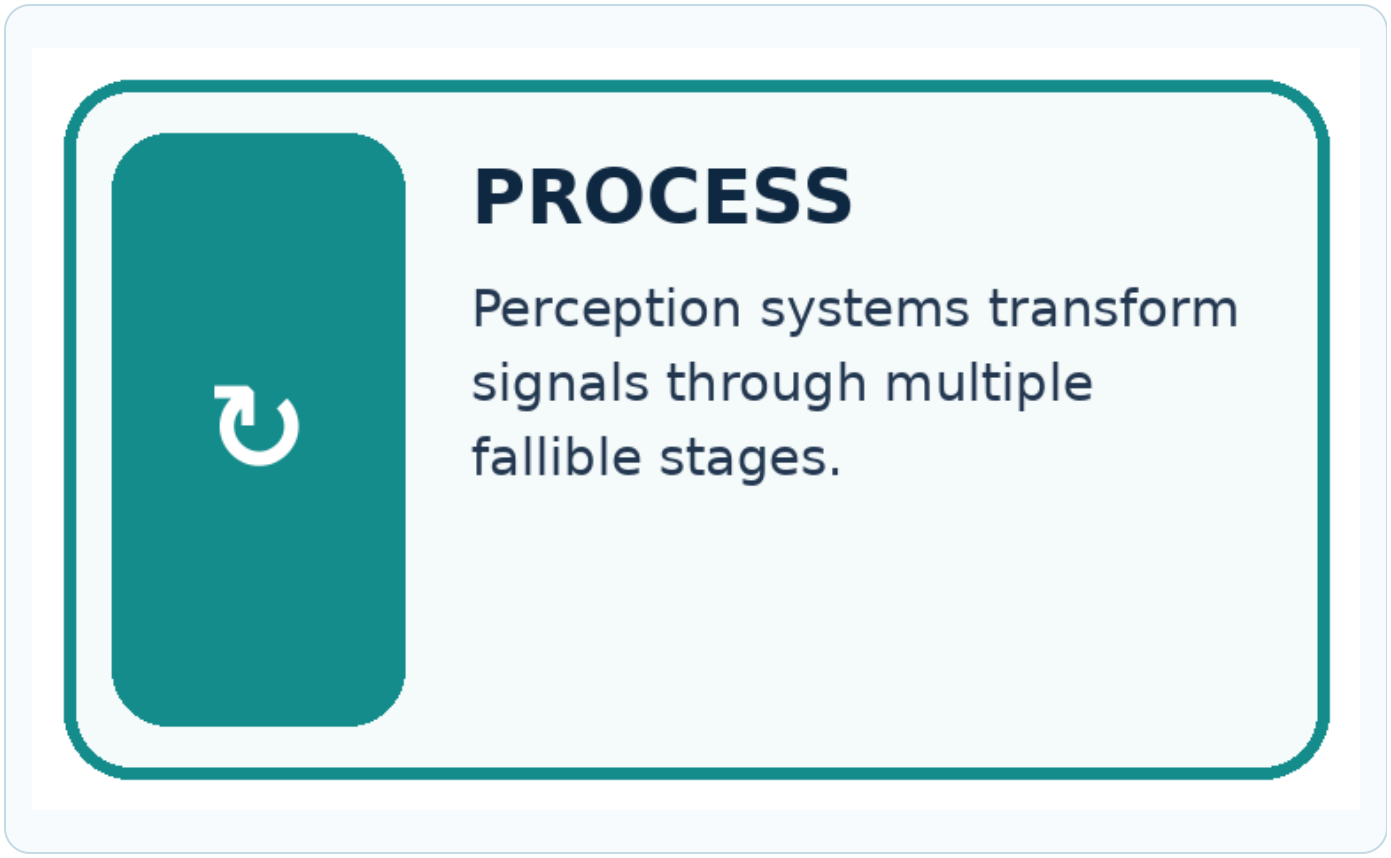
VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Perception systems transform signals through multiple fallible stages.
OUTPUT
OUT
Trace speech-to-command and image-to-label pipelines.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare conflicting **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

Speech pipelines and multimodal evidence — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.	Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.	There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

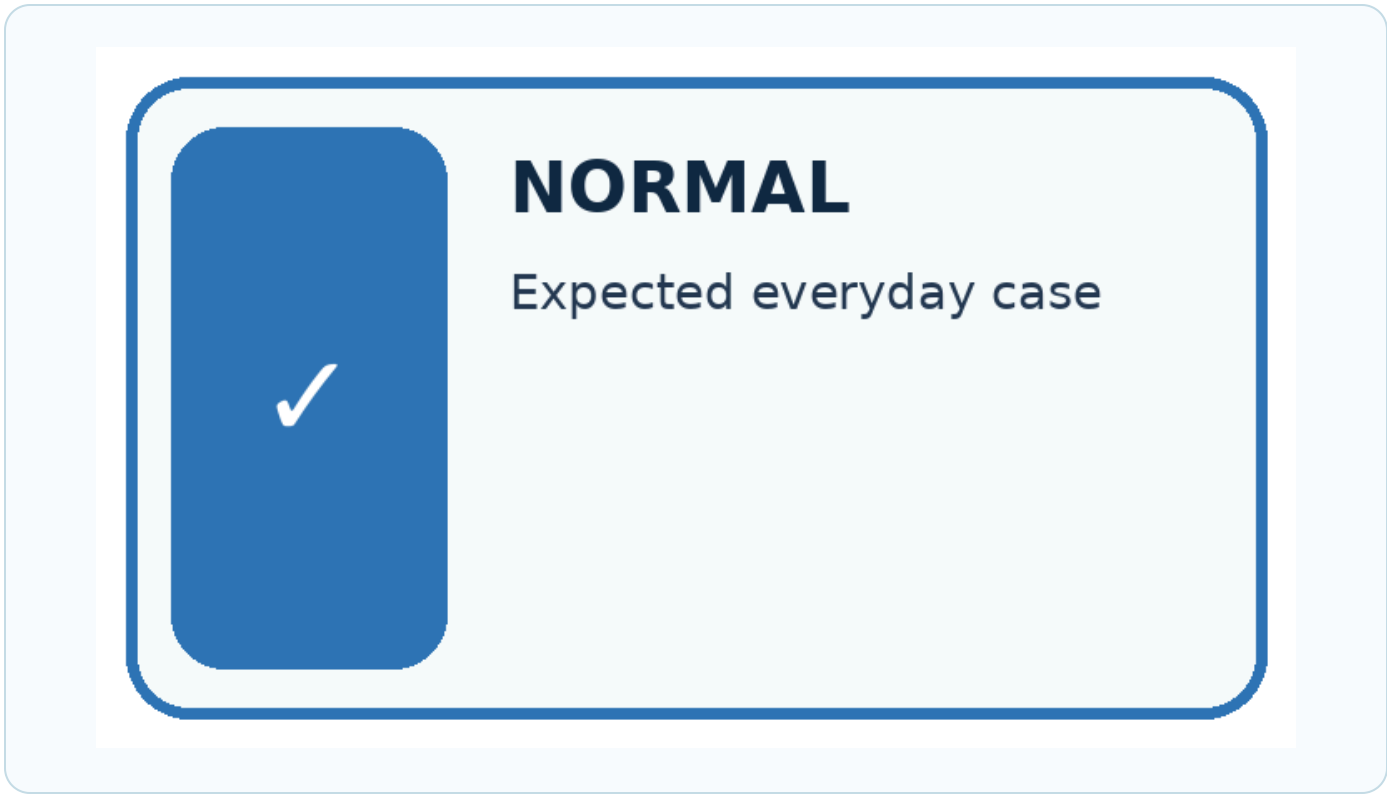
> Compare the supplied cases for Image and Voice Pipelines. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

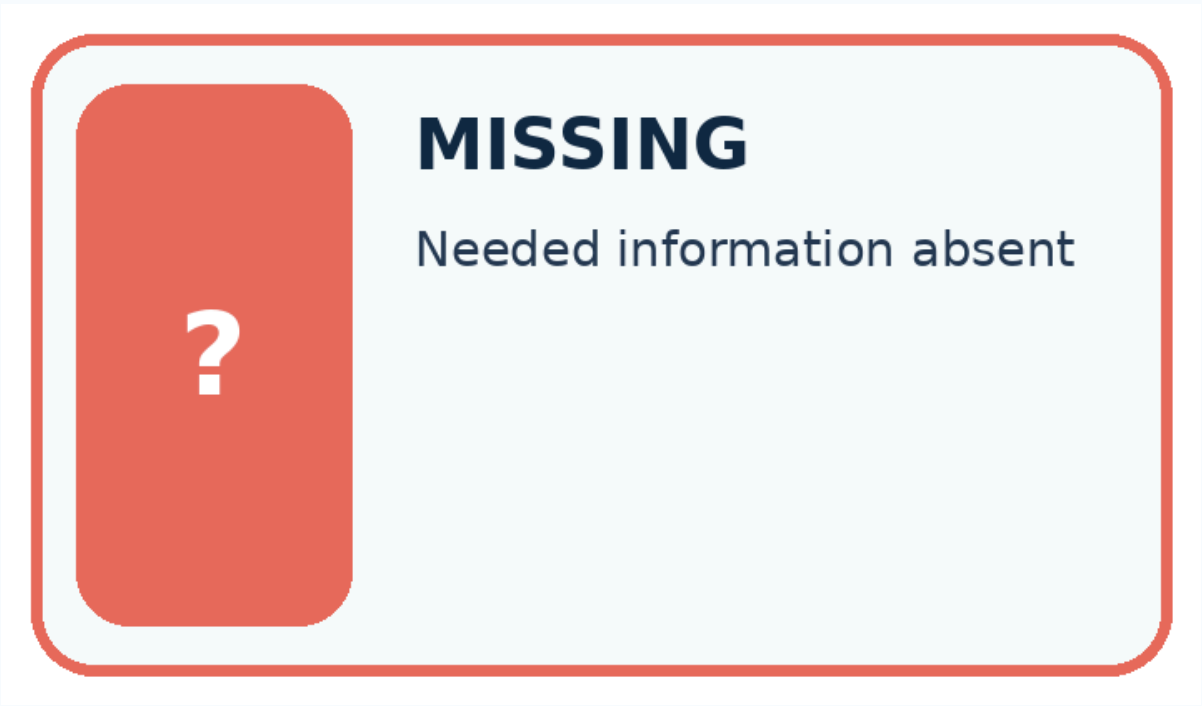
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Remove access to the audio, while keeping the authoritative inventory.

Predicted result

Prediction: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.

Why the result follows

Reason: The inventory resolves the shelf task; it does not recreate unavailable audio.

STUDENT LESSON · 5 OF 18

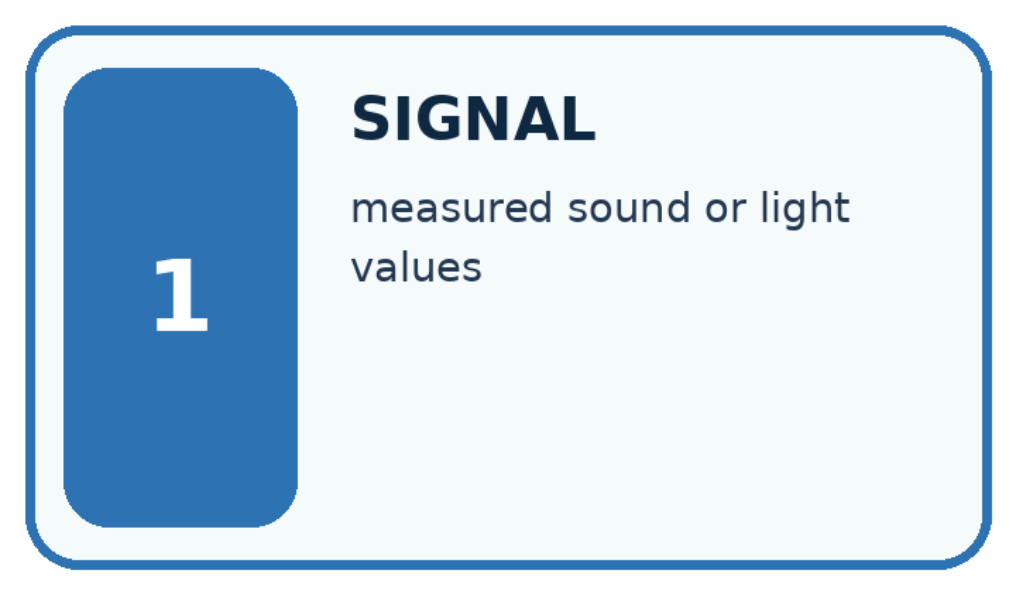
Words that unlock the idea

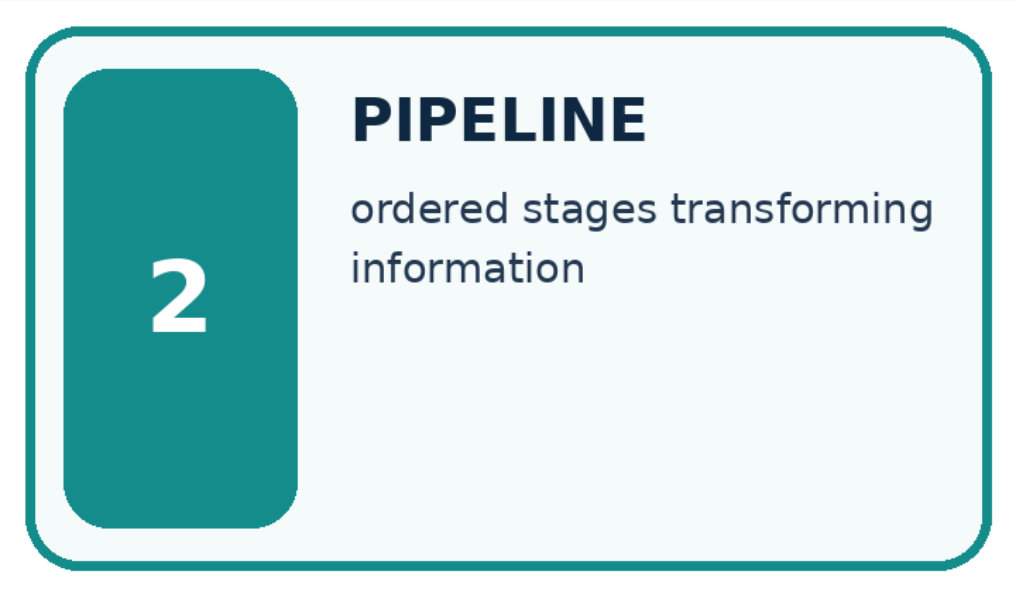
> Use the vocabulary of Image and Voice Pipelines to explain a concrete decision, rather than only memorising definitions.

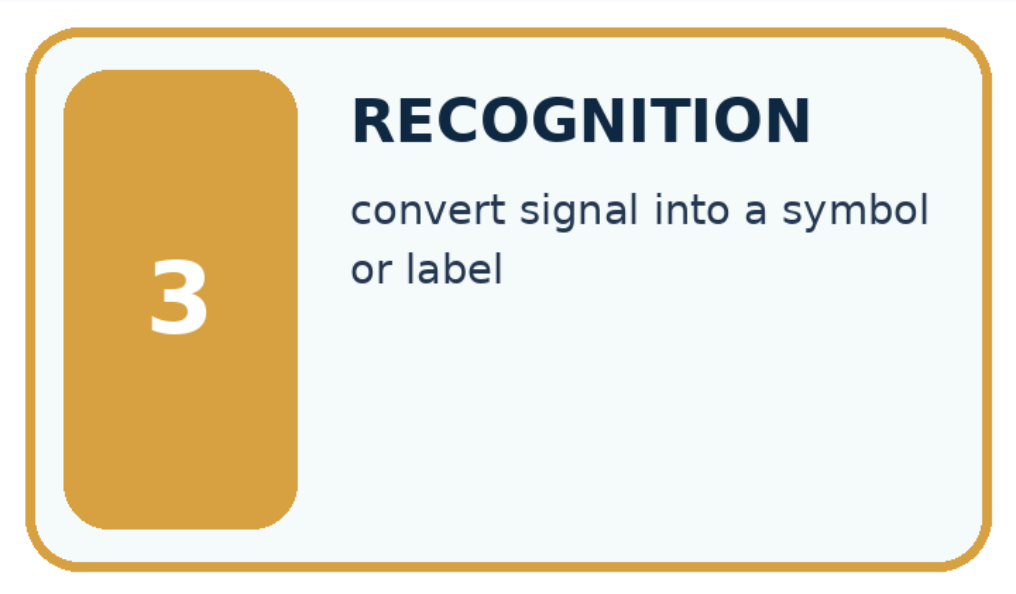
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
SIGNAL
measured sound or light values


2
PIPELINE
ordered stages transforming information





Word	My own example	Not this / boundary
signal	A measurable quantity carrying information, such as changing sound pressure or light intensity.	
pipeline	A connected sequence of stages in which one stage's output becomes another's input.	

Word	My own example	Not this / boundary
recognition	Interpreting an input as a known category or pattern rather than merely recording its signal.	
stage error	A failure introduced at a particular step of a pipeline.	

Words used in the closer explanation

Word	Meaning
Modality	A form of information such as audio, image or text
Transcription	A text representation of speech
Error propagation	An earlier error affecting later processing

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Image and Voice Pipelines, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 26 • WORKED EVIDENCE

1. Voice: sound → sampled signal → recognized words → intent → action.
2. Image: pixels → features → score → label.
3. Noisy sound can produce wrong words even if the intent stage works.
4. Diagnose the first wrong stage before changing later stages.

1. Voice: sound → sampled **signal** → recognized words → intent → action.

2. Image: pixels → features → score → label.

3. Noisy sound can produce wrong words even if the intent stage works.

4. Diagnose the first wrong stage before changing later stages.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Worked explanation · Image and Voice Pipelines

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Image and Voice Pipelines

A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.

The facts we will use

Supplied facts

Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

Supplied facts

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.

AI ACADEMY / AI-2/26

Image and Voice Pipelines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Audio input

Known phrase names fourteen, not forty.



2. Transcription

Text output changes the shelf number to forty.



3. Cross-check

The authoritative inventory gives shelf 14.



4. Review and correct

Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

CHANGE ONE THING

Remove access to the audio, while keeping the authoritative inventory.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Audio input	Known phrase names fourteen, not forty.
2. Transcription	Text output changes the shelf number to forty.
3. Cross-check	The authoritative inventory gives shelf 14.
4. Review and correct	Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A correct later lookup cannot repair an earlier wrong **input** by magic. Locate the first difference between known audio and transcript, then compare the stated source authority.

Keeping both records makes the correction auditable. A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

What this example does not establish

Keep the stated limit

Declared authority resolves this task's source choice; it does not make every inventory entry universally accurate.

Keep the stated limit

Read the labelled picture: Which stage first fails, and why should the faulty transcript not be silently overwritten?

Keep the stated limit

Change one thing: Remove access to the audio, while keeping the authoritative inventory. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Image and Voice Pipelines claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: The final model is the only possible source of a wrong output. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

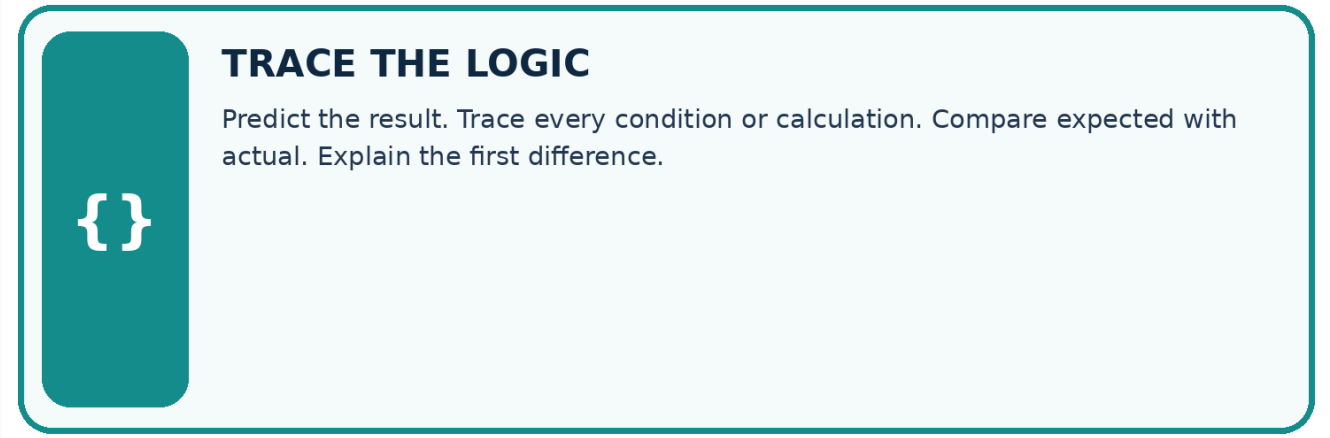
Practise together

Practise Image and Voice Pipelines with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Speech **pipeline**s and multimodal **evidence**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Speech **pipeline**s and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which stage first fails, and why should the faulty transcript not be silently overwritten?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

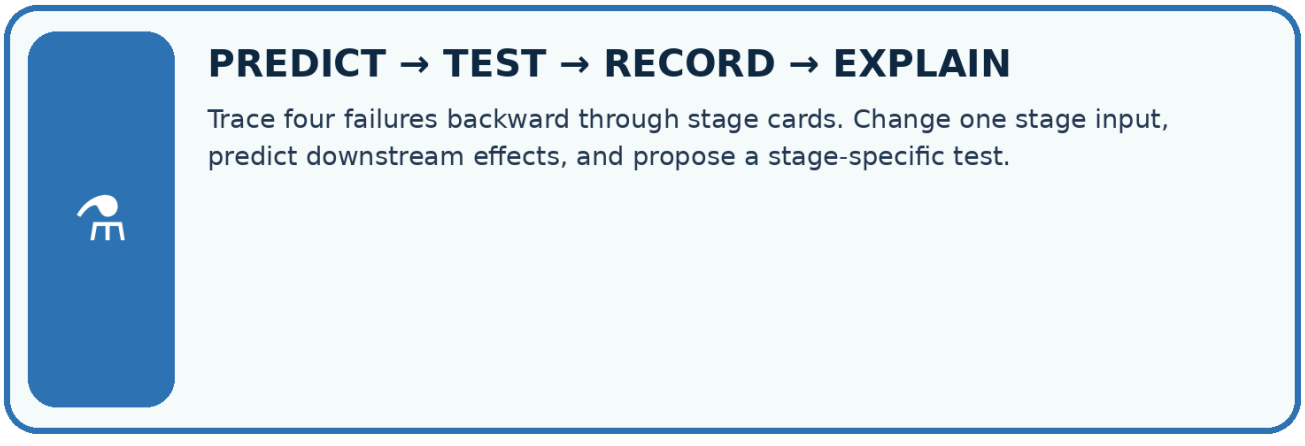
Run the investigation

> Investigate Image and Voice Pipelines. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Trace four failures backward through stage cards. Change one stage **input**, predict downstream effects, and propose a stage-specific test.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

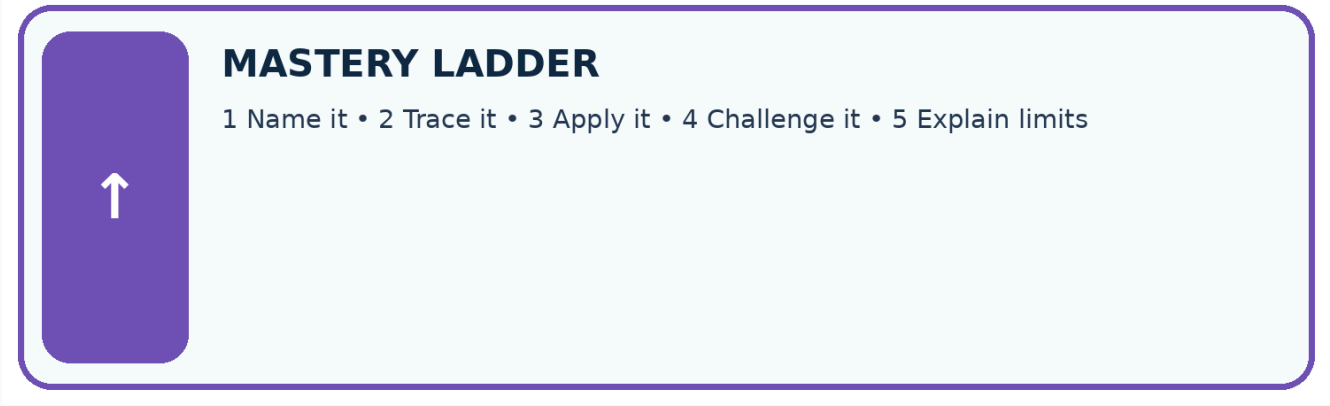
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Image and Voice Pipelines. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

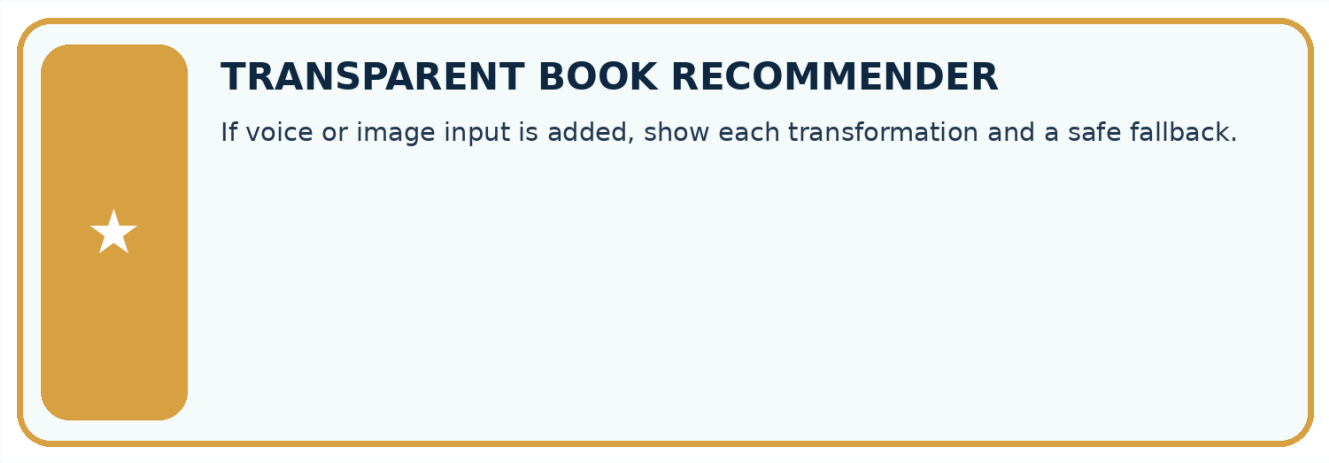
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT If voice or image **input** is added, show each transformation and a safe fallback.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance.

STUDENT LESSON · 12 OF 18

Show what you know

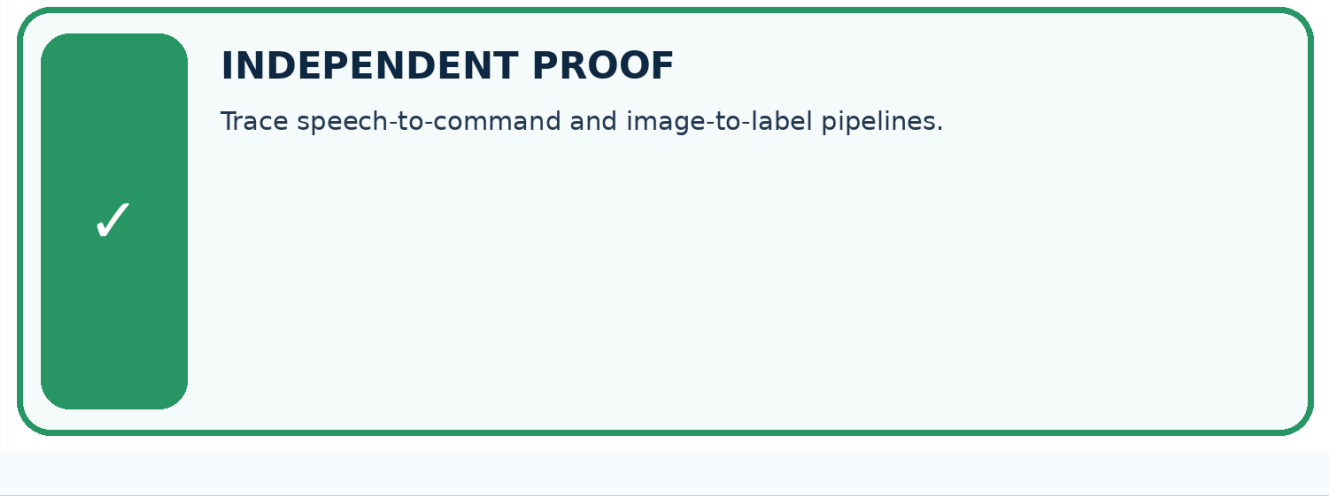
> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Trace speech-to-command and image-to-label **pipeline**s.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Image and Voice Pipelines case for supported practice. Repeated examples do not become fresh assessments.

Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A voice book helper transcribes speech, extracts preference, then ranks books.

The question

> Where can a wrong suggestion begin?

Worked explanation

Transcription can mishear poetry and pass the wrong preference downstream. Check intermediate results.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Locate the first error in a voice pipeline

Evidence or stage	Value or result
Speech	Reader says poetry
Transcript	May mishear the preference
Extract	Uses the transcript's genre
Rank	Wrong input can produce a wrong suggestion

A voice book helper transcribes speech, extracts preference, then ranks books.

> Where can a wrong suggestion begin?

Transcription can mishear poetry and pass the wrong preference downstream. Check intermediate results.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 26 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A voice helper hears find a book, converts speech to text, then searches. It writes find a cook. Which stage first failed and what should you inspect?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

If voice or image **input** is added, show each transformation and a safe fallback.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A voice request says 'under 90 pages'. The transcript says 'under 19 pages'; filtering correctly finds no books. Identify the first failing stage and a useful confirmation step.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Image and Voice Pipelines, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.

By the end, I can	How I will show it
Explain the mechanism and terminology of speech pipelines and multimodal evidence	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Image and Voice Pipelines without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of speech **pipeline**s and multimodal **evidence**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

1. Audio **input**

2. Review and correct

3. **Transcription**

4. Cross-check

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Review and correct Replay the audio, preserve the faulty transcript and use confirmed shelf 14. A correct later lookup cannot repair an earlier wrong **input** by magic.

Locate the first difference between known audio and transcript, then compare the stated source authority. Keeping both records makes the correction auditable.

A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove access to the audio, while keeping the authoritative inventory. Which conclusion is supported by the supplied facts?

1. Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.

The inventory resolves the shelf task; it does not recreate unavailable audio.

Repair a missing building block

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

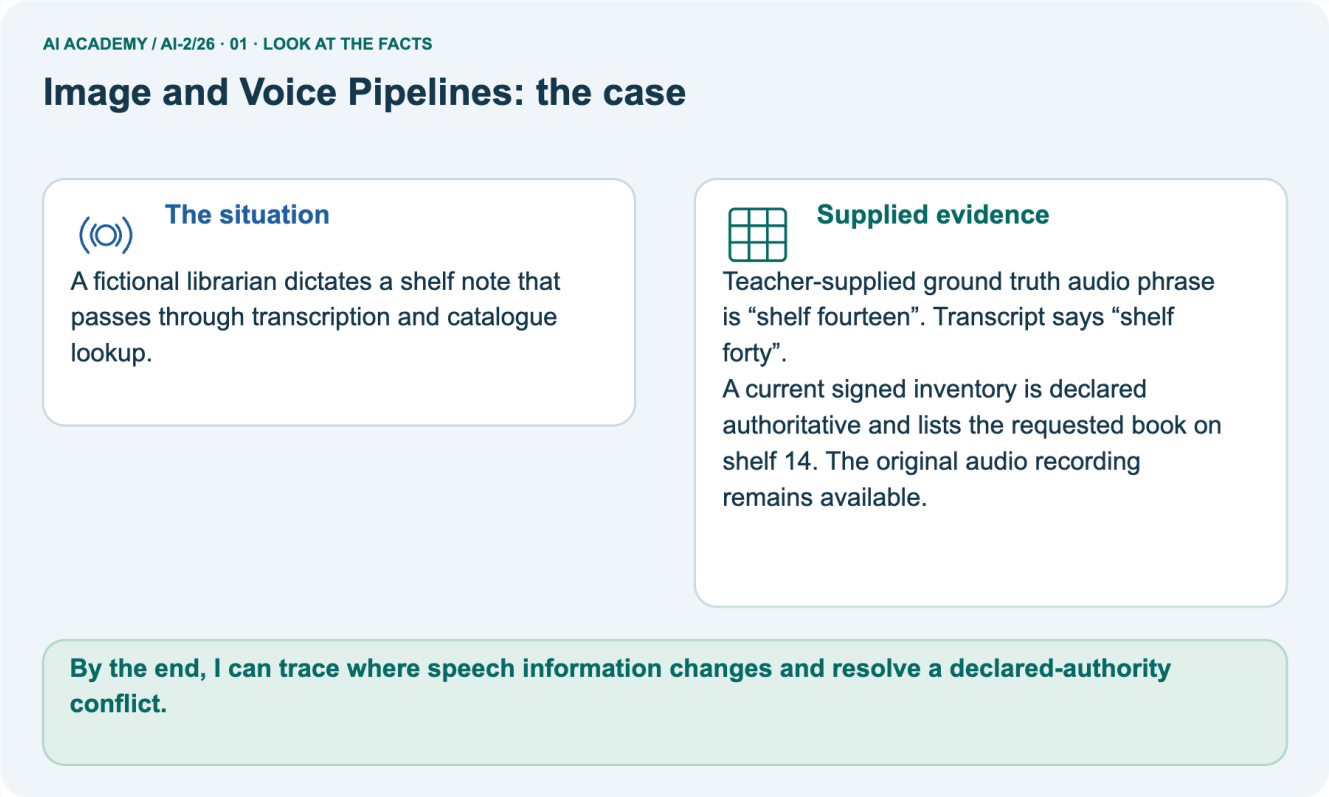
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Image and Voice Pipelines graphic using its labels, deciding evidence and relevant limitation.

01. Image and Voice Pipelines: the case



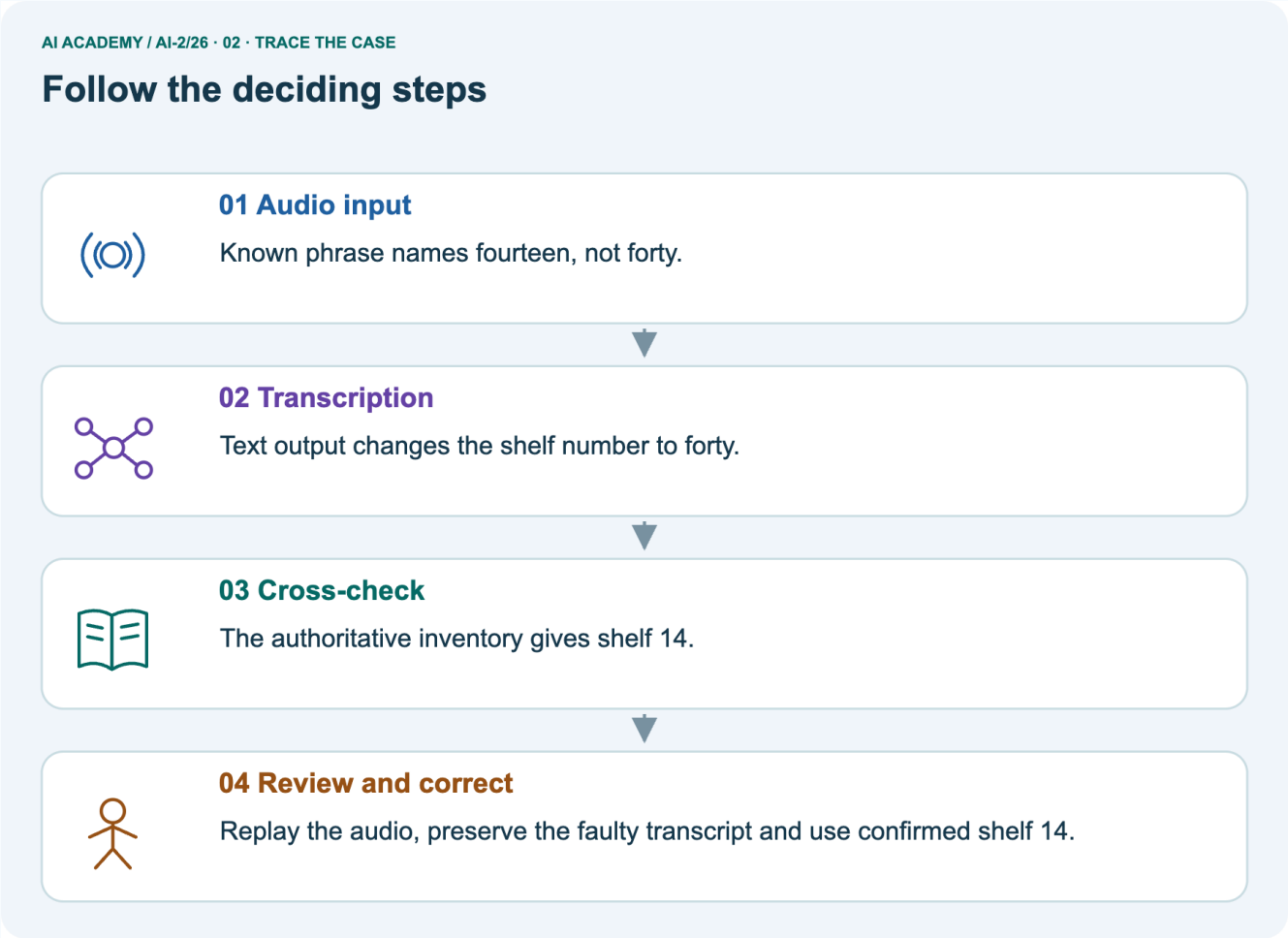
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup. Supplied facts: Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

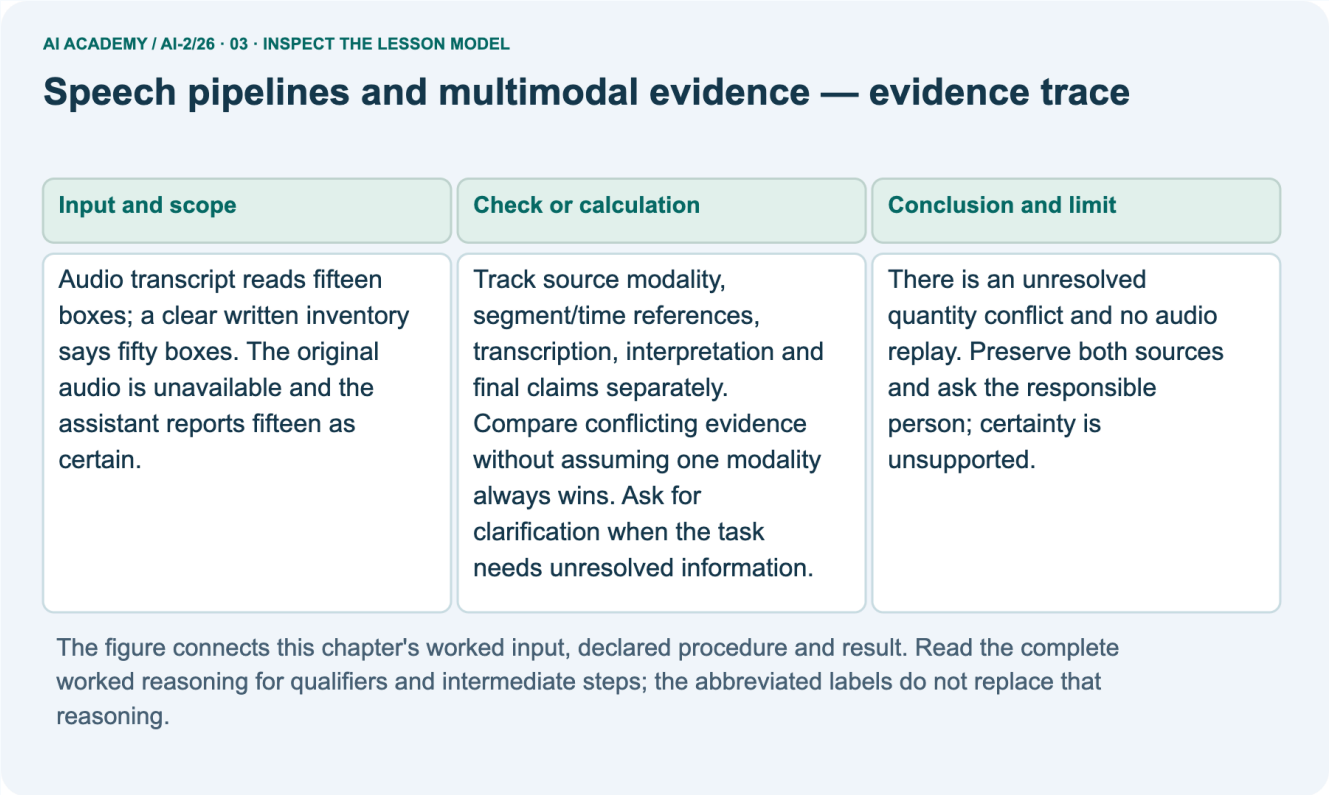


Follow the labelled picture with the matching explanation. Open the vector diagram

Audio **input**: Known phrase names fourteen, not forty. → **Transcription**: Text **output** changes the shelf number to forty. → Cross-check: The authoritative inventory gives shelf 14. → Review and correct: Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech **pipeline**s and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

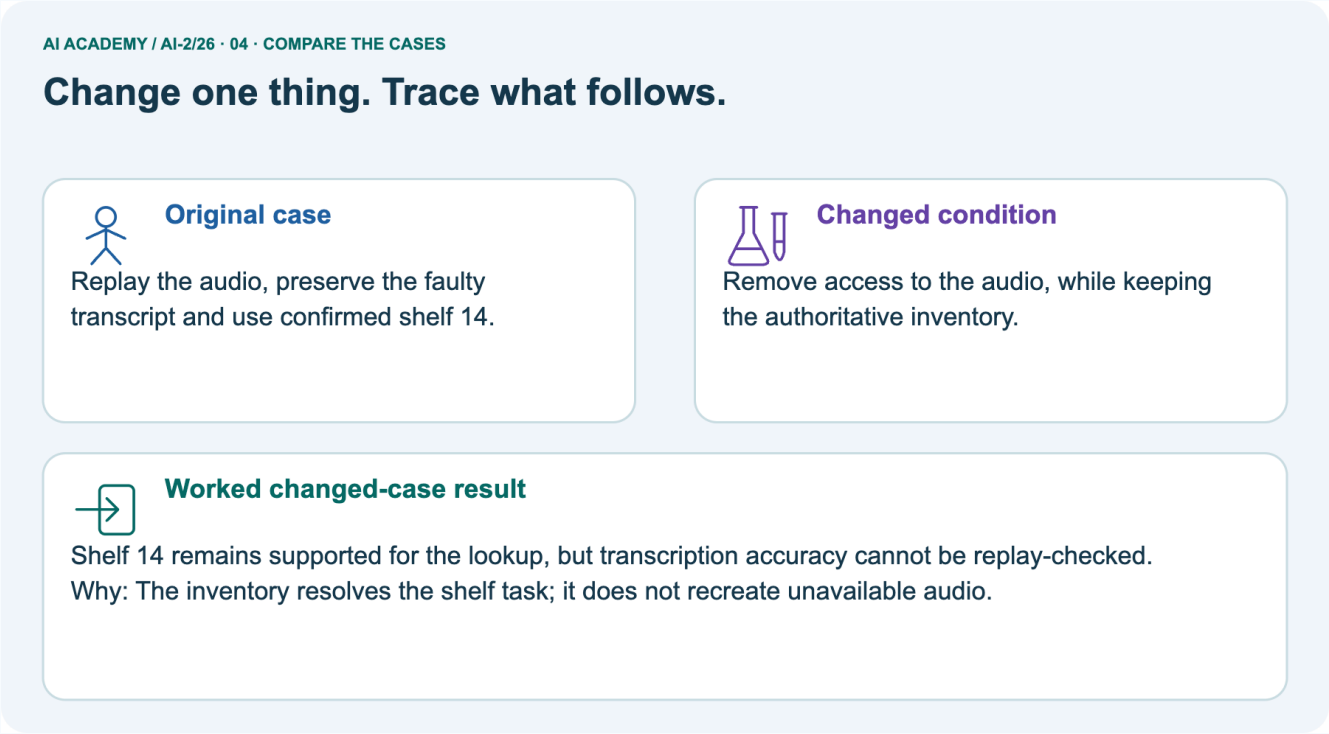
The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity conflict and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



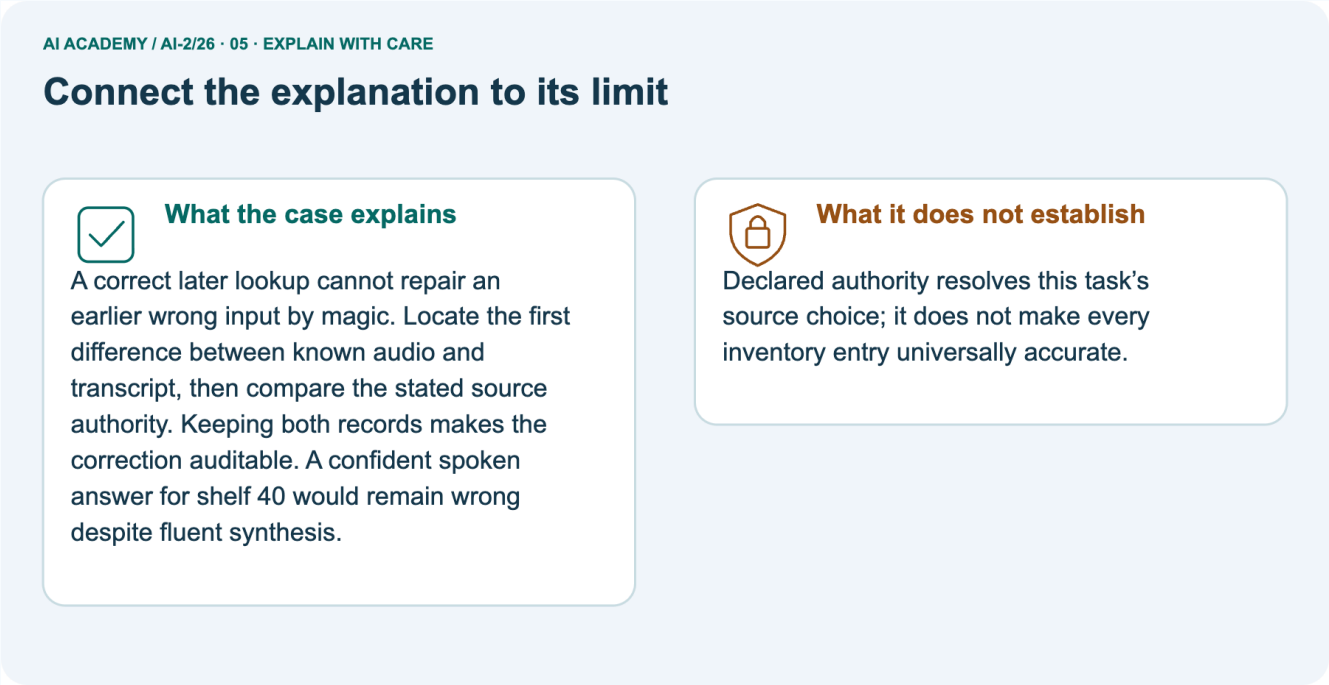
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Replay the audio, preserve the faulty transcript and use confirmed shelf 14. Changed condition: Remove access to the audio, while keeping the authoritative inventory.

Expected result: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked. Reason: The inventory resolves the shelf task; it does not recreate unavailable audio.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

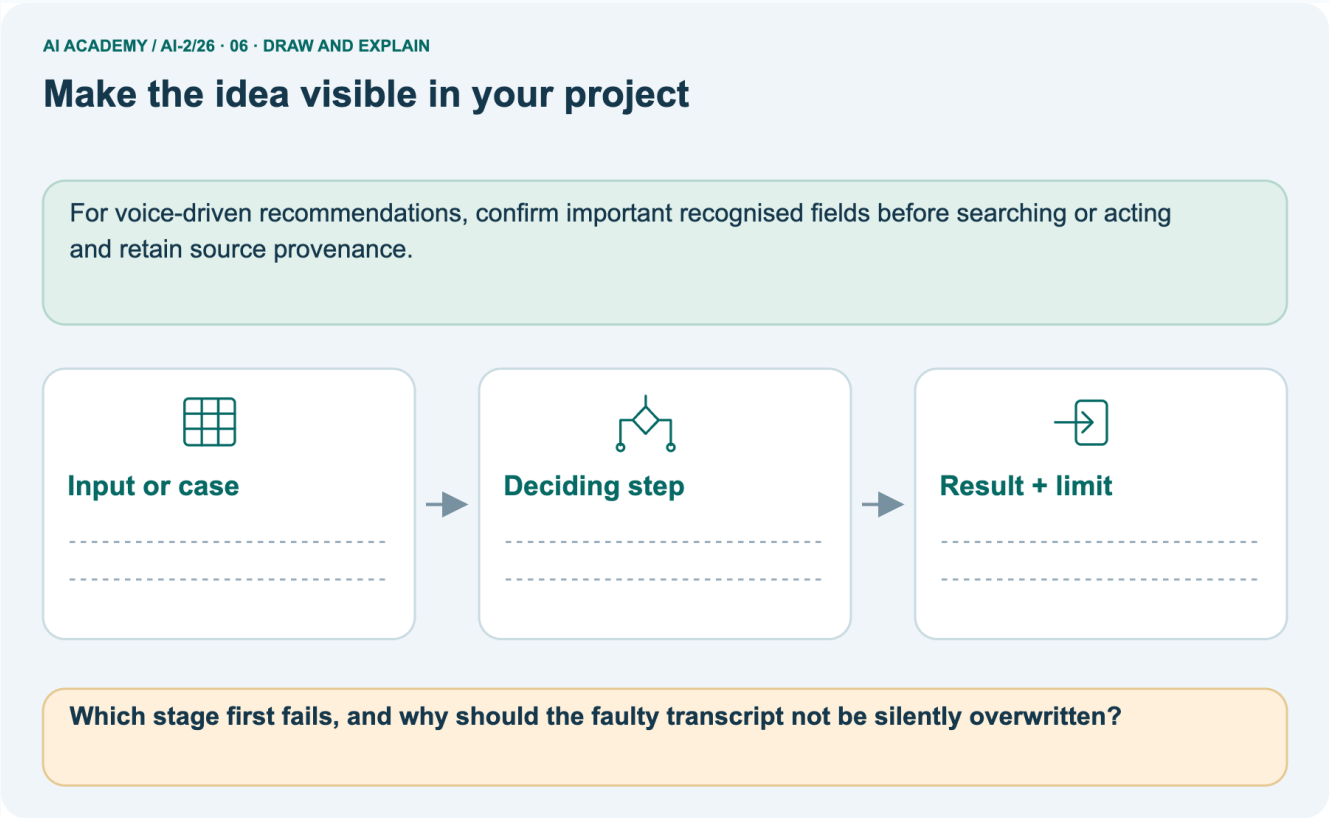
A correct later lookup cannot repair an earlier wrong **input** by magic. Locate the first difference between known audio and transcript, then compare the stated source authority.

Keeping both records makes the correction auditable. A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Limit: Declared authority resolves this task's source choice; it does not make every inventory entry universally accurate.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance. Draw your **input**, deciding step, result and limit.

Supported question: Which stage first fails, and why should the faulty transcript not be silently overwritten?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.